Agentic Graph Reasoning: Multi-Hop Knowledge Graph Question Answering with Structural Claim Verification

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***Abstract*—Large language models answer factual questions fluently. But they sound just as fluent when they do not hold the facts. Retrieval-augmented generation grounds answers in retrieved text, but retrieves once, before reasoning starts. Systems that navigate a knowledge graph do better by interleaving retrieval with reasoning. Even they send out whatever their final generation call produces, and nothing checks the answer's claims before it reaches the user. We propose Agentic Graph Reasoning (AGR), a knowledge-graph question-answering framework whose agent plans sub-objectives and walks the graph through a deterministic tool API. Before the answer is emitted, a Structural Verification Layer splits the draft into atomic claims and checks each one against the triples the agent traversed. We evaluate AGR on 400 certified questions from each of WebQSP and ComplexWebQuestions, against four baselines sharing one frozen backbone and one call budget. AGR reaches a Hits@1 of 0.755 on WebQSP and 0.522 on ComplexWebQuestions, beating every baseline at half the calls of the agentic baseline. That baseline leads on the questions it finishes, so the margin is affordability under a fixed budget. On ComplexWebQuestions, AGR is the only system whose accuracy rises with hop count. An ablation finds one detectable effect, opposite to what we expected. Removing the planner improves WebQSP accuracy. The verification layer does not change how often AGR is right. It makes answers auditable. A hand reading of all 256 remaining failures names the *echo attractor*, a grounded entity next to the answer that several systems fall into together.**

***Index Terms*—Knowledge graphs, question answering, large language models, retrieval-augmented generation, hallucination, claim verification, agents.**

I. Introduction

LARGE language models answer factual questions fluently. But often enough they give wrong answers, and those errors are not random. They are systematic, and the model states them with the same confidence it gives a correct answer [1]. So a fabricated answer looks exactly like a correct one. Things get worse when the answer requires putting several facts together. A *multi-hop* question such as "Who directed the film that won Best Picture in 1998?" needs a chain of lookups in a fixed order. If the first hop lands on the wrong entity, the rest of the chain carries on from a false premise. And a check on the final answer alone never sees the intermediate results.

Hallucinations arise at three points, and each calls for a different kind of intervention [1]. Defects in the training data can be fixed only by curating the corpus and retraining. The model itself stores what it knows diffusely across its weights, with no reliable signal that separates a memorised fact from a plausible continuation. The third point is the prompt, and it is the one this work takes on. When the context supplied with a question already carries the facts needed to answer it, the model does not have to fall back on parametric recall. Retrieval augmentation rests on that premise, and so does this work. Our contribution is in how that context is assembled, and in what gets checked before the answer goes out.

Retrieval-augmented generation (RAG) [2] attaches the model to an external corpus and lets it answer from retrieved passages. For a multi-hop question, the ranking signal measures the wrong thing. The passage holding the intermediate entity may bear little resemblance to the question. And flat retrieval cannot represent the relationships that a compositional question has to traverse [3]. A knowledge graph makes those relationships explicit, so composing two facts becomes a traversal. GraphRAG [4] exploits this, but it still retrieves *statically*. It fetches a neighbourhood once, before reasoning starts. A fixed radius is then either too small to contain the answer or so large that the relevant edges are buried. The natural response is to interleave retrieval with reasoning. Think-on-Graph [5] set that pattern with model-guided beam search over the graph. Later systems added decomposition, backtracking, and generated relation paths [6], [7].

Agentic navigation is the right shape for the problem, and this paper adopts it. Three things it leaves unsolved motivate the work.

Nothing checks what the final answer asserts. Agentic systems ground their answers in the subgraph they traverse, and the entities they assert do exist in the graph. The problem is not fabrication but *precision*. An entity can sit in the retrieved subgraph, have been retrieved correctly, and still be the wrong answer. Every system we survey emits whatever its final generation call produces from the traversed context. Post-hoc self-checking reduces factual error in free-form generation [8]. What none provides is a claim-level check against the retrieved triples, carried out *before* the answer is emitted, inside a graph-navigating agent.

The contribution of individual mechanisms is unquantified. Agentic systems are assemblies of decomposition, scoring, backtracking, and self-correction. The literature tends to report them as wholes. So when one beats a static baseline, the mechanism that produced the gain is rarely established.

Accuracy is reported without cost. Every exploration step costs at least one model call. An accuracy table that leaves

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cost out makes agentic methods look better across the board. The honest picture is a trade-off.

We present Agentic Graph Reasoning (AGR), a knowledge-graph question-answering framework built as an explicit state machine over a constrained, deterministic graph tool API. Its distinguishing component is a *Structural Verification Layer*. Before anything is emitted, the layer splits the draft answer into atomic claims. It checks each one against the (head, relation, tail) triples the agent actually traversed. Claims it cannot ground are re-explored or withdrawn. So where it cannot ground a claim, AGR hedges rather than asserts. And the answers it does give come paired with the triples that support them.

Stated formally, the input is a natural-language question q and a knowledge graph $G = (E, R, T)$ with entities E, relation types R, and triples $T \subseteq E \times R \times E$. The system returns an answer set $A \subseteq E$ together with a supporting triple set $S \subseteq T$. The set S contains only triples the system traversed in reaching A, and it is checked before A is emitted. Three properties set this apart from standard question answering over knowledge graphs. The output is a pair, and its supporting triples are part of the contract. That is what makes the reasoning auditable rather than merely asserted. The support must also be traversed, which rules out justifying, after the fact, an answer the system reached some other way. And the justification is checked before the answer is emitted. So unsupported content can be dropped or hedged instead of merely flagged. Three research questions follow.

- **RQ1:** Does agentic navigation improve multi-hop factual accuracy, and does the advantage grow with the number of hops?
- **RQ2:** What does pre-generation verification contribute beyond graph navigation?
- **RQ3:** Which components contribute what, at what token cost?

We answer them with a controlled evaluation. All five systems share one frozen backbone, one knowledge environment, and one call budget. So any difference between them comes from architecture, not model capacity. The paper makes four contributions.

1) AGR and its Structural Verification Layer (Section IV). The word *structural* names what the check establishes and, in the same word, bounds it. The check asks whether an edge joins a claim's endpoints, not whether it is the edge the claim names.
2) A controlled comparison of five systems on 400 certified questions from each of WebQSP [9] and ComplexWebQuestions (CWQ) [10] (Sections V and VI). It splits the margin over the agentic baseline into search quality and affordability under a fixed budget.
3) A component-level ablation with paired significance tests (Section VI-G). It finds that decomposition is stratum dependent. Removing the planner improves accuracy on the shallower benchmark and trends toward harming it on the deeper one.
4) A hand reading of all 256 remaining failures (Section VII). It names the *echo attractor*, a recurring failure in which several systems sharing a graph and a backbone converge on the same grounded but wrong answer. It also counts the benchmark's own label defects, which adjudication separates from those convergences.

The headline results are these. AGR leads every accuracy column on both datasets, and all eight paired tests against the baselines reject. Where the shared call cap lets the agentic baseline finish, that baseline is slightly ahead. So the margin is affordability, not better resolution. Structural hallucination is eliminated by graph navigation itself, since the agentic baseline also reaches zero. And removing the verification layer changes the correctness of two questions out of 398. Its contribution is auditability rather than accuracy.

One limit applies to every absolute number. The knowledge environment is a union of per-question subgraphs, each pre-extracted to contain its own question's answer. That makes it friendlier than full Freebase [11], and the accuracies are not comparable with evaluations over full Freebase. The comparisons between systems are sound, because every system navigates the same environment (Section VIII-D).

The rest of the paper is organised as follows. Section II reviews related work, and Section III describes the knowledge environment. Section IV specifies AGR. Section V gives the experimental setup and Section VI the results. Section VII reads the failures. Section VIII discusses the findings and their limits, and Section IX concludes.

II. Related Work

A. Static Retrieval over Text and Graphs

Retrieval-augmented generation [2] gives a language model material retrieved at the moment it answers. The answer can then rest on that evidence rather than on *parametric* knowledge, the facts absorbed into the model's weights. Dense retrieval encodes queries and documents into a shared vector space [12], and sentence encoders produce embeddings suited to direct cosine comparison [13]. Graph-based retrieval swaps flat passages for structured context. GraphRAG [4] builds a graph over a corpus and answers from summaries of its communities. HippoRAG [14] ranks nodes by a random walk that restarts from the query's entities. Other systems learn the retriever, scoring candidates with a graph neural network [15] or retrieving a bounded subgraph with a light-weight scorer [16]. All of them retrieve once, before reasoning, so the context cannot be conditioned on what reasoning finds. The distinction that matters here is not graph versus vector but *static versus interleaved*.

B. Agentic Exploration of Knowledge Graphs

Think-on-Graph [5] established *training-free* agentic question answering over knowledge graphs, in which the model is prompted and never fine-tuned. The model performs iterative beam search over the graph, with beam width and depth both set to three, and prunes relations and entities itself. Its successor adds document retrieval at each step [17]. Plan-on-Graph [6] decomposes the question into sub-objectives. Its reflection step decides both whether to self-correct and which entity to backtrack to, which makes it the one surveyed system with

genuine backtracking. That self-correction operates on the search, not on the draft answer. Reasoning-on-Graphs (RoG) [7] fine-tunes a model to generate relation paths and instantiates them in the graph. Its answers are faithful by construction, and graph-constrained reasoning [18] takes the same route. But the plan is generated once, with no mechanism to revise it when it fails to instantiate. KG-Agent [19] drives a toolbox with an instruction-tuned model. StructGPT [20] reads structured data through specialised interfaces. Generate-on-Graph (GoG) [21] admits model-generated triples when the graph is incomplete, whereas AGR admits only graph triples.

KBQA-o1 [22] is the system closest to genuine tree search. It wraps a ReAct-style [23] agent in full Monte Carlo Tree Search [24], [25], with policy and reward models, rollouts, and value backpropagation. AGR has none of these. Its search is guided best-first search with a bounded beam and backtracking [26], [27], and each candidate is scored once. KBQA-o1 also shows what an ablation can attribute. Removing its tree search drops its F1 on GrailQA [28] from 78.5 to 48.5 [22]. That attribution was possible only because its authors removed one component at a time.

C. Verification and Factuality

Chain-of-Verification [8] has the model draft a response, plan verification questions, answer them, and regenerate. Multi-agent debate [29] gets a related effect by having several model instances critique one another. Both check against the model's own knowledge, and models cannot reliably correct their own reasoning without outside feedback [30]. A second group critiques the retrieval. Self-RAG [31] emits reflection tokens that judge whether a passage supports the output. Corrective RAG [32] grades the retrieved documents and triggers a fallback search. Reflexion [33] conditions an agent's next attempt on a critique of its failed one, and a self-correcting graph RAG system applies such a loop to clinical decision support [34]. FActScore [35] breaks generated text into atomic facts and checks each against a reference source. It is a metric rather than a control step, and the groundedness metric of Section V-F borrows its decompose-then-verify protocol.

D. Positioning

Table I compares the closest systems on the mechanisms that matter here. Decomposition is well established. Backtracking exists in Plan-on-Graph, and RoG is faithful by construction. But no surveyed system decomposes its draft answer into atomic claims and checks each one against the retrieved triples before responding. That slot is the one AGR fills.

Published accuracies are not used to rank systems in this paper, and Table II lists them for orientation only. Backbones differ across papers, some papers evaluate on sampled subsets, and answer-matching conventions vary. The same methods drop sharply when run on smaller open-weight backbones, and RoG and KG-Agent are fine-tuned on these benchmarks. Think-on-Graph's own paper reports two variants that differ by 1.8 and 1.9 points on CWQ with the same two backbones [5]. That gap is the order of many claimed improvements in this literature. So the agentic baseline here is reimplemented and run under controlled conditions (Section V-E). The one published comparison is reported with its differences stated (Section VI-C).

TABLE I
MECHANISMS OF THE CLOSEST PRIOR SYSTEMS. NO SURVEYED SYSTEM CHECKS THE CLAIMS OF ITS FINAL ANSWER AGAINST RETRIEVED TRIPLES BEFORE RESPONDING.

System	Exploration	Decom-poses	Backtracks	Verifies before answering
Think-on-Graph	Model-pruned beam search	No	No	No
Plan-on-Graph	Adaptive planning with reflection	Yes	Yes	The search, not the answer
RoG	Fine-tuned relation-path plans	Partly	No	Faithful by construction
KG-Agent	Fine-tuned tool selection	Implicitly	No	No
GoG	Search with generated triples	Yes	No	No
StructGPT	Interfaces read in turn	No	No	No
KBQA-o1	Tree search over an agent	Implicitly	By tree revisits	No
AGR (ours)	Guided best-first beam search	Yes	Yes	Yes, claim by claim

TABLE II
HITS@1 AS EACH PAPER PUBLISHES IT. THESE FIGURES ARE NOT COMPARABLE WITH ONE ANOTHER OR WITH THIS PAPER. BACKBONES AND PROTOCOLS DIFFER, AND TWO OF THE SYSTEMS ARE FINE-TUNED.

System	Backbone	WebQSP	CWQ
Think-on-Graph [5]	GPT-3.5	76.2	57.1
	GPT-4	82.6	67.6
Plan-on-Graph [6]	GPT-3.5	82.0	63.2
	GPT-4	87.3	75.0
RoG [7]	Fine-tuned LLaMA2-7B	85.7	62.6
KG-Agent [19]	Fine-tuned 7B	83.3	72.2
GoG [21]	GPT-3.5	78.7	55.7
	GPT-4	84.4	75.2
StructGPT [20]	GPT-3.5	72.6	—

III. THE KNOWLEDGE ENVIRONMENT

Everything the agent can know, it knows from the graph. So the environment sets a ceiling on how accurate any system can be. A result reported without that ceiling is hard to interpret.

A. Freebase and Mediator Nodes

A knowledge graph $G = (E, R, T)$ stores facts as triples (h, r, t), each asserting that relation r holds from head entity h to tail entity t. Triples are atomic, so a claim can be checked by testing for specific edges. They are also composable, and a chain of them expresses a multi-hop relationship that no single triple states. Freebase [11], frozen since 2016, is the substrate of the standard benchmarks. One of its features changes every depth measurement. A *mediator node* is an unnamed node standing for one fact with more than two participants, with one edge to each participant. That Rainn Wilson played Dwight Schrute in *The Office* is stored as a mediator with edges to the actor, the character, and the show. So a relationship a user would describe as one hop is two edges in the graph. Distances in this paper are in *raw* hops, which count graph edges. One *logical* hop through a mediator is two raw hops.

B. Hallucination, Defined Operationally

"Hallucination" means different things in different papers. This paper adopts a narrow, operational definition tied to the environment. An *asserted* entity is one the system names in its answer. It is *ungrounded* if it does not exist in the graph, or exists but lies more than four hops from every topic entity of the question. A *claim* is a single statement drawn from the answer that relates two entities. It is *unsupported* if the traversed triples do not entail it. Two consequences follow. The definition is relative to the environment, so an entity absent from the graph is ungrounded even if it is true in the world. And groundedness is not correctness. An answer can be grounded in every entity and still be the wrong answer to the question.

C. Construction

The environment is the union of every per-question subgraph that the Reasoning-on-Graphs distribution [7] publishes for WebQSP and ComplexWebQuestions, across all splits. So the questions and the graph come from one distribution and are mutually consistent. Taking the union is not a minor detail. A per-question subgraph on its own is close to an oracle. It was extracted to contain the answer, so an agent searching it meets almost no distractors. Merging every question's neighbourhood into one graph brings the distractors back. Table III records the result. Mediators are retained rather than collapsed, since systems that skip them lose the facts they encode [21]. They make up 63.7% of the nodes.

Two consequences must be disclosed. The subgraphs were pre-extracted to guarantee that answers are reachable, which makes the environment friendlier than raw Freebase. And the distribution keys entities by name rather than by Freebase identifier. So distinct entities that share a surface form are merged into one node. Both bound the absolute numbers, and Section VIII-D returns to them.

The graph is stored in Neo4j [36], with each relation as a native relationship type, and is queried in Cypher [37]. Entity names and the relation vocabulary are embedded with the 384-dimensional sentence encoder all-MiniLM-L6-v2 [13]. Relation names are verbalised before embedding, so people.person.place_of_birth is encoded as the phrase it means. Embeddings have a narrow role. They link question phrases to entities and pre-rank relation candidates. They never establish that a fact is true, a role that belongs to the triples alone.

Entity resolution is a three-stage cascade of exact, lexical, and vector matching, each stage tried only if the previous one fails. Over all $7{,}149$ topic-entity mentions in both test sets, 99.86% resolve at the exact stage. The ten that do not are bare machine identifiers that the benchmark carries in place of a name. So a failure in this environment cannot be blamed on the resolver missing the question's topic. The graph and the benchmarks share their name strings, though, so this shows only that the resolver loses nothing the environment contains.

D. The Validation Gate

No experiment ran until the environment was certified. For every test question we computed whether a gold answer is reachable from a topic entity within four raw hops, by breadth-first search over undirected edges. The agent's tools look in both directions, so the gate does the same. Reachability is 97.3% on WebQSP and 99.7% on CWQ (Table III). It is the accuracy ceiling, since no navigating system can find an answer it cannot reach. Existence and reachability agree question for question. When a question is unanswerable here, it is because its gold entities are absent, not because they are too far away. Reachability was also re-verified directly in Cypher, independently of the code that computed it.

TABLE III
THE KNOWLEDGE ENVIRONMENT AND ITS VALIDATION GATE. REACHABILITY IS THE SHARE OF TEST QUESTIONS WITH A GOLD ANSWER WITHIN FOUR RAW HOPS OF A TOPIC ENTITY. IT IS THE HITS@1 CEILING OVER EACH FULL TEST SPLIT.

Property	Value
Entities	2,592,892
Triples	8,309,194
Distinct relation types	7,058
Mediator nodes	1,652,618 (63.7%)
Mentions resolved by exact match	7,139 of 7,149 (99.86%)
Reachable, WebQSP test split	1,584 of 1,628 (97.3%)
Reachable, CWQ test split	3,519 of 3,531 (99.7%)

Certifying reachability says nothing about answers that are not entities. The import gives every node a single label, with no typed literal. Only 77 nodes carry a full date as their name and 175 a bare year, together under 0.01% of the graph and untyped. And the tool layer exposes no sorting or comparison operator. So "the earliest" or "the smallest" cannot be computed even when every candidate is present. These gaps define a class of questions that are *unanswerable in this environment*, those whose answer is a date, a measured quantity, or a superlative selection. Missing one is not a reasoning failure, and Section VII counts how often it happens.

IV. THE AGR FRAMEWORK

A. The Controlled-Agent Pattern

There are two ways to build a graph-navigating agent. The first hands the model a set of tools through native function calling and lets it drive the loop itself, in the style of ReAct [23]. The second inverts control. The *program* orchestrates the loop, calls the graph tools deterministically, and consults the model only at decision points, through constrained prompts that must return JSON. AGR uses the second. Every graph operation is then logged by construction, and budgets are enforced in code rather than requested in prompts. A malformed model response can spoil one decision, but it cannot corrupt the control flow. The agency in AGR lives in the decision policy. That policy decides which sub-objectives to pursue, which edges look promising, whether a path is a dead end, and whether the draft answer is supported.

B. The State Machine

AGR is a state machine of six nodes over a shared typed state, built with LangGraph (Fig. 1). The topology has cycles. Exploration loops until the evaluator is satisfied, and verification

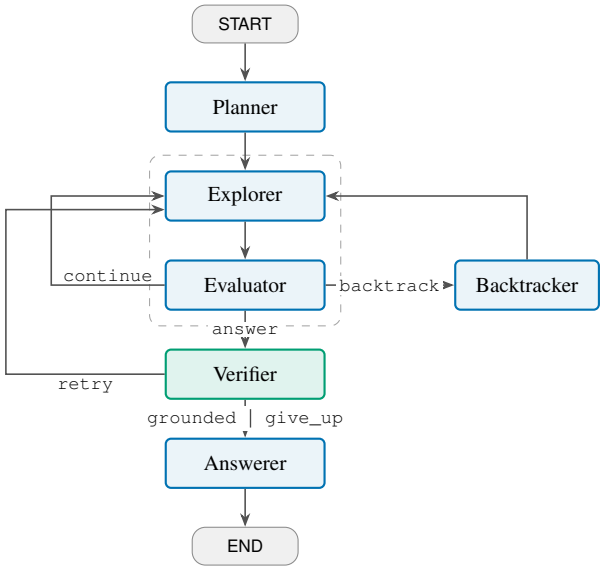


Fig. 1. The AGR state machine. The dashed region is the explorer–evaluator search loop. The backtracker restores an earlier frontier, and the verifier's retry edge is verification-driven re-exploration. All three cycles are bounded by the budgets of Table IV. Every path from the evaluator to the answerer passes through the verifier.

can send the agent back to explore. The verifier sits between the decision to answer and the answer itself. There is no path from the evaluator to the answerer that does not pass through verification.

C. The Graph Tool API

The model never writes graph queries. A generated query introduces a failure that is indistinguishable, in the results, from a reasoning failure. When a question is answered wrongly, one cannot tell whether the agent reasoned badly or emitted invalid syntax. So the tools are parameterised Cypher templates, written once and unit-tested. Four operations are live.

search_entity links a mention to candidate nodes through the resolver of Section III. It is the only entry point from text to graph.

get_relations returns the relation types incident to an entity in *both* directions, each with its fanout count. Freebase's edge direction is a modelling artefact. And the count is free signal, since a relation with a fanout of forty thousand rarely answers a specific question. Relations in administrative namespaces such as common, type, and freebase are dropped. The list is capped at 300 relations, sorted by fanout.

get_neighbors expands an entity along one relation. When it reaches a mediator, it hops through it and returns the entities on the far side, each with the chain of relations that reached it. So the model reasons about logical neighbours, while the log keeps the raw chain. At most 200 neighbours are returned per expansion.

verify_connection asks whether two entities are adjacent in the full graph, directly or through one mediator. It ignores the relation and the direction. A fifth operation, verify_triple, also matched the relation. It was built first as the claim checker and did not survive contact with generated text. Claim decomposition produces relations phrased in English, while Freebase predicates are neither phrased that way nor reliably one edge long. So requiring the relation to match rejected claims that were true, and no node calls it.

D. Planner, Explorer, and Scorer

The planner makes one model call. It turns the question into an ordered chain of at most four sub-objectives, in which later objectives refer back to earlier results with a #N back-reference. The prompt is few-shot, with one example for each of four archetypes (single hop, composition, conjunction, and superlative). The single-hop example shows that *not* decomposing is a valid output. The plan is validated rather than trusted. If it is empty, too long, refers forwards, or names no topic, the planner falls back to a single sub-objective holding the whole question. Planning can degrade but can never abort a run. Search is seeded from the dataset's annotated topic entities, the standard protocol in this literature [5], [7]. That choice separates reasoning from entity linking.

The explorer performs one expansion of the frontier. It pushes a checkpoint onto the backtrack stack and gathers the relations of every anchor into a single candidate list. It scores the whole list in one pass and takes the best three edges per anchor, skipping banned ones. Each selected edge is expanded, and the new frontier is sorted by the score of the edge that produced each entity and truncated to five anchors. Each candidate edge c receives the hybrid score

$$\mathrm{score}(c) = \alpha \cos\left(\mathbf{v}_{\mathrm{rel}(c)}, \mathbf{v}_{\mathrm{obj}}\right) + (1-\alpha)\,\mathrm{LLM}(c), \quad (1)$$

where $\mathbf{v}_{\mathrm{rel}(c)}$ embeds the candidate's relation and $\mathbf{v}_{\mathrm{obj}}$ embeds the active sub-objective. $\mathrm{LLM}(c) \in [0,1]$ is the model's rating of how likely following the edge is to advance that objective. Three properties matter. The objective is embedded, not the question, with any back-reference already replaced by the resolved entity's name. That substitution is the mechanism by which decomposition is supposed to help. The model rates only the twenty candidates that the embedding ranks highest, in one batched call. Every other candidate scores $\mathrm{LLM}(c) = 0$, so the pre-filter is a hard gate. And the blend is applied after the call. The scoring prompt holds no configuration value, so every condition of an α sweep shares the same cached model responses. We use $\alpha = 0.7$, chosen on development data (Section V-G).

E. Evaluator and Backtracking

The evaluator makes one model call per expansion. It sees the active sub-objective, the thirty traversed facts most relevant to it, and the remaining budget. It decides whether to continue, backtrack, or answer, and it names the entities that satisfy the objective. Those names are not taken on trust. Any name that is not the tail of a traversed triple is discarded. A language model asked which entities satisfy an objective will happily list ones it knows from training but never retrieved. This *groundedness filter* refuses to let that recall pass as evidence. On one question of an early smoke run, the evaluator listed nine correct countries that appeared in no retrieved triple, because the beam had gone down an unrelated branch. The filter rejected all nine, and the

Algorithm 1 AGR navigation, with the verification layer abstracted

Require: question q, budgets, blend α, threshold τ
Ensure: answer A with supporting triples S

$P \leftarrow \textsc{Plan}(q)$; **if** invalid **then** $P \leftarrow [\,q\,]$
$F \leftarrow \textsc{SeedAnchors}(q)$; $T, C, stack, ban \leftarrow \emptyset$
while budgets permit **do**
  push $(F, depth, \sigma)$ onto $stack$
  $R \leftarrow \bigcup_{e \in F} \textsc{GetRelations}(e)$
  score R by Eq. (1) against the active objective
  $E \leftarrow$ best 3 edges per anchor, excluding ban
  $T \leftarrow T \cup \textsc{GetNeighbors}(E)$
  $F \leftarrow$ best 5 new entities by edge score
  $\sigma \leftarrow$ the largest of the three signals of this pass
  $(d, done, \rho) \leftarrow \textsc{Evaluate}(\text{objective}, \textsc{TopFacts}(T))$
  $\rho \leftarrow \rho \cap \textsc{Tails}(T)$; $C \leftarrow C \cup \rho$ {groundedness filter}
  if $done$ **then**
    advance P; **if** no objective remains **then** $d \leftarrow$ answer **else** $F \leftarrow \rho$
  end if
  if no new triples **or** $\sigma < \tau$ **or** $d =$ backtrack **then**
    $ban \leftarrow ban \cup E$; restore the highest-scoring snapshot
  else if $d =$ answer **then**
    break
  end if
end while
$(A, S) \leftarrow \textsc{VerifyAndAnswer}(q, T, C)$ {Algorithm 2}
return (A, S)

system abstained. That cost a point, but it kept every proposed entity tied to the graph. Surviving resolutions are added to the candidate answers on every evaluation, whether or not the objective was judged complete. When an objective completes, the plan advances and the resolved entities become the next frontier.

Backtracking fires on the first of three triggers, checked in this order. Either the last expansion produced no new triples, or the frontier's signal fell below a threshold $\tau = 0.20$, or the evaluator judged the direction a dead end. The first two cost nothing, so a mechanically hopeless frontier is abandoned without asking the model. The signal compared against τ is the largest of three quantities. They are the best blended score among the expanded edges, the best raw embedding similarity over every candidate, and the best model rating. So the trigger fires only when all three are weak. The backtracker restores the *highest-scoring* snapshot on the stack rather than the most recent one. It also bans the (anchor, relation, direction) combinations of the last pass. Without that ban list, a deterministic scorer facing a restored frontier would select the same edges again until the budget ran out. Algorithm 1 states the whole navigation loop.

The implementation departs from this specification in one way that has a measurable consequence. It is recorded rather than repaired, since repairing it would change the frozen results. The backtracker bans the *most recent* pass but restores the *highest-scoring* snapshot, which is often older. So whatever was expanded in between stays unbanned, and the explorer can select it again. Over the main test runs, 53 explorer passes expanded a set of (anchor, relation) pairs that the same question had already expanded. That is exactly the repetition the ban list exists to prevent.

TABLE IV
BUDGET CONFIGURATION. THE LAST TWO ROWS ARE CAPS IN THE TOOL LAYER THAT BOUND WHAT THE AGENT CAN SEE.

Budget	Value	What it bounds
max_depth	4	explorer expansions
beam_width	3	edges expanded per anchor
max_anchors	5	frontier size
max_backtracks	3	backtracks per question
max_llm_calls	25	model calls per question
max_verify_iters	2	repair iterations
max_seconds	300	wall-clock time
max_relations	300	relations offered to the scorer
max_fanout	200	neighbours per expansion

F. Budgets and Termination

Table IV lists the budgets. They are enforced in the routers between nodes and in the model client, never in prompts. The depth budget counts explorer expansions. One expansion can cross a mediator, so it does not bound the search at four raw hops. Every cycle in Fig. 1 passes through a router that checks a counter which only moves one way. So no sequence of model outputs can produce a run that never ends. Every model call is also cached, keyed on the model, its sampling settings, and the full prompt. A cache hit replays the original token counts, so a cached rerun reproduces the original run's budget record exactly. Each run writes two logs. The tool log records every graph operation with its arguments, result size, and latency. The run record holds the answer, the draft, the verification outcome, every backtrack with its trigger, the full decision trace, and the budget snapshot. It also names the backbone, the budget configuration hash, the run configuration, and the source commit. So any record found anywhere states what produced it.

G. Design Validation on a Smoke Set

Twenty questions from the training and validation splits, never from test, were used once to check the assembled framework's mechanics. The set was built to exercise specific machinery rather than to sample the benchmarks. The first end-to-end run answered roughly three or four of them correctly, and the final configuration answers thirteen or fourteen. That interval is where several of the mechanisms above were shown to be necessary. The first run's failures collapsed into five systematic causes, and Table V gives the mechanism each one motivated. Its before-state was read from a console log that was never committed. So these figures are given as ranges, and none of them enters any result.

The two confabulations of the first run came from a retrieval defect, not a generation defect. Two questions about national currencies produced a confident "United States Dollar". The relation cap had hidden the currency relation, the beam surfaced triples about national income per capita in dollars, and the answerer matched the pattern. No retrieved triple supported the answer, and nothing that examined the answer caught it. After the cap was raised, both questions resolved correctly at roughly a third of the token cost. But a claim-level check against the traversed triples would have caught them regardless

TABLE V
ROOT CAUSES OF FAILURE IN THE FIRST END-TO-END RUN ON THE SMOKE SET, AND THE MECHANISM EACH ONE MOTIVATED.

Observed failure	Mechanism introduced
Backtracking was a no-op, since a restored frontier reproduced the same scores and decisions	A ban list on abandoned (anchor, relation, direction) combinations
The fact window showed the thirty most recent triples, so a large expansion pushed the answer out of view	A fact window ranked by relevance to the objective
A relation cap of 80, sorted by fanout, hid precise low-fanout relations behind hub relations	The cap raised to 300, and administrative predicates blocked
Entities resolved while an objective was incomplete were discarded	Candidates accumulated on every evaluation
The frontier was truncated in insertion order, so the agent drifted from the anchor that mattered	The frontier sorted by edge score before truncation

of the retrieval defect. That is the argument for verifying before emitting.

H. The Structural Verification Layer

The navigation loop ends with a set of traversed triples and a list of candidate entities. Nothing in it constrains what the final answer *asserts*. A final generation call is free to name an entity that appeared in neither, as the currency answers of Section IV-G did.

The response is to check the draft before it is emitted. Chain-of-Verification and FActScore decompose and check after generation, and both take further model output as evidence [8], [35]. The layer borrows their decomposition step and rejects their placement and their evidence. Placing the check inside the loop has three consequences. The checker sees the same evidence the drafter saw. A rejected claim can trigger more exploration while budget remains. And the judge of whether a fact holds is the environment, not a second opinion from the model that wrote the draft.

Claim decomposition. One model call produces three things at once. They are a prose draft, the entities the draft asserts as the answer, and a decomposition of the draft into atomic claims $\langle h, r, t\rangle$ over entity names. The call sees at most sixty traversed triples, ranked by relevance to the question, and up to twenty-five candidates. The prompt requires names to be copied exactly from the facts, and it requires a hedged draft to carry no claims. It also requires the answer entities to be specific entities of the kind the question asks for. That rule was added after drafts such as "the senators from New Jersey are the United States Senate members associated with New Jersey". Such an answer is grammatically complete and semantically empty. The relation slot r is free text, since the model has no reliable way to name a Freebase predicate. Decomposition is sparse. On the development set the drafter emitted 121 claims over 80 questions, with a median of one per question. Ten questions produced no claims at all, so for those ten the layer certified nothing.

Algorithm 2 Structural verification of a draft

Require: question q, traversed triples T, candidates C
Ensure: draft, answer entities A, claim sets S and U, evidence E

$W \leftarrow \textsc{TopFacts}(q, T, 60)$
$\mu \leftarrow$ name-to-id map over T
$(\mathit{draft}, A, K) \leftarrow \textsc{DraftAndDecompose}(q, W, C)$
$\mathit{adj} \leftarrow \{(h,t),(t,h) : (h,r,t) \in T\}$
$S \leftarrow \emptyset$; $P \leftarrow \emptyset$; $E \leftarrow \langle\,\rangle$
for all $c = \langle h, r, t\rangle \in K$ **do**
  if $h \notin \mu$ **or** $t \notin \mu$ **then**
    $P \leftarrow P \cup \{c\}$ {never traversed}
  else if $(\mu[h], \mu[t]) \in \mathit{adj}$ **then**
    $S \leftarrow S \cup \{c\}$; append the joining triples to E
  else if verify_connection$(\mu[h], \mu[t])$ **then**
    $S \leftarrow S \cup \{c\}$ {no evidence recorded}
  else
    $P \leftarrow P \cup \{c\}$
  end if
end for
$U \leftarrow \emptyset$
if $P \neq \emptyset$ **then**
  split P into S and U by $\textsc{Entail}(W, P)$
end if
if $U \neq \emptyset$ **and** repair budget remains **then**
  add a repair objective for the first claim in U
end if
route grounded if $U = \emptyset$, else retry if the repair, depth, and time budgets remain, else give_up
return $(\mathit{draft}, A, S, U, E)$

Two-stage checking. Algorithm 2 gives the procedure, and Fig. 2 draws one claim's path through it. The structural check first maps a claim's names back to nodes through the traversed triples alone. A claim naming an entity the agent never traversed cannot reach either structural route. It falls to the entailment check. The first structural route is *traversed adjacency*. The claim is supported if a traversed triple joins its endpoints, and every such triple is attached to the answer as evidence. The second route asks verify_connection whether the endpoints are adjacent in the full graph, since the agent may have reached both along different branches. It records no evidence. This second route is rare. On the test sets it was consulted for 100 of the 2,110 claims and accepted 39. Claims that fail both routes go to the *entailment check*, one batched model call that judges each claim against the same fact window. If that call fails for any reason, every pending claim is marked unsupported.

The structural check runs first for three reasons. The graph should decide where it can, so the model is consulted only where the graph is silent. Only traversed adjacency yields triples that can be attached to the answer. And a set lookup is free, while the entailment call is the layer's only unavoidable cost beyond drafting.

Repair. The router after the verifier has three outcomes. With no unsupported claim it routes to grounded, and the draft is emitted with no further model call. With unsupported claims and the repair, depth, and time budgets all remaining, it routes to retry. The verifier has already appended a repair objective for the first unsupported claim and re-anchored the search on that claim's head entity. Anything else routes to give_up. On that path the final entity list is computed deterministically.

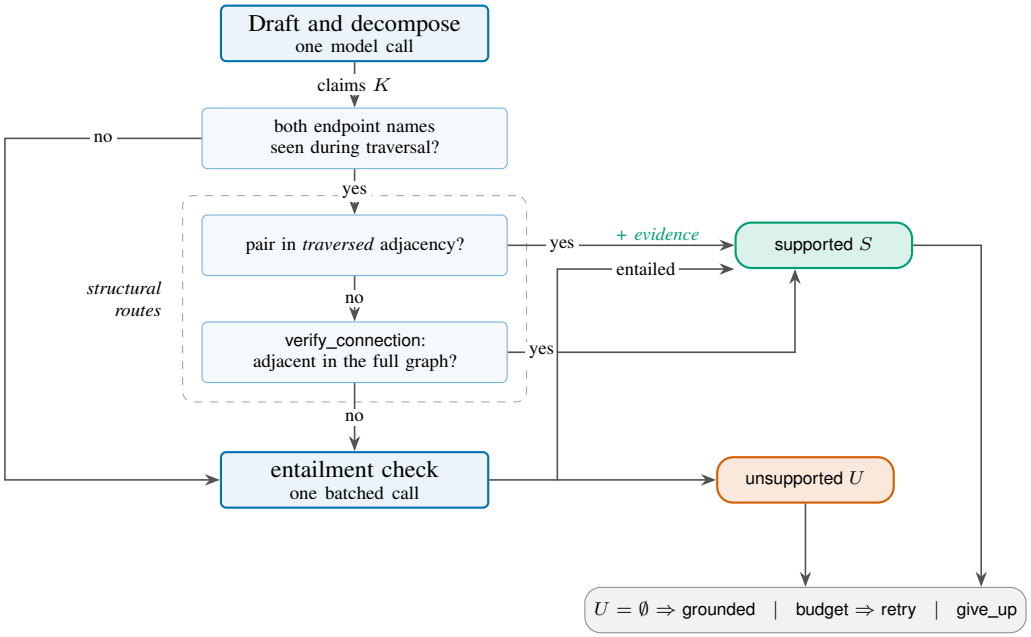


Fig. 2. The path of one atomic claim through the verification layer, mirroring Algorithm 2. A claim leaves at the first test that settles it. Only the first structural route adds to the evidence, which is why some grounded answers carry supporting triples and others do not. A claim that names an entity the agent never traversed skips both structural routes, because the name map is built from traversed triples alone.

An entity is kept if it appears in a supported claim or in no unsupported claim. One rewrite call then restates the prose around the kept list. Repair is capped at two iterations. The grounded path costs nothing, so the layer is cheap overall. On the development set it raised mean tokens per question from 4,858 to 4,925, about 1.4%.

The retry route never ran on the development set. On all three development questions that produced an unsupported claim, the depth budget was already spent when the verifier was reached. That correlation does not survive the move to test. Across the 800 test questions the verifier asked for a repair on 52, and exploration actually resumed on 28 of them. So the depth condition blocks the retry about half the time. Thirteen questions ended grounded after a repair, and the iteration cap bound on 15.

That deterministic filter replaced a second generation call, and the reason is the most instructive result of the design. On the development set, a question-by-question comparison against the same system with claim checking switched off found exactly three questions that differed. No entailment verdict was involved in any of them. All three traced to the rewrite call, which at the time regenerated the answer entities along with the prose. In one case it dropped four correct entities together with an unsupported one. In another it added the question's own subject to the answer. In the third it respelled the graph's University Yale as Yale University, which scored as wrong (Section IV-I). The verification logic was sound, and its repair path frequently was not. Replacing generation with filtering removed two of the three regressions.

The output contract and its limits. AGR returns each answer paired with the triples that support it. That coverage is narrower than the phrase suggests. Only the traversed-adjacency route records evidence. And the run log keeps the number of supporting triples, not the triples. So the contract holds inside the run, but it cannot be checked afterwards from the committed records. On the development set, 13 of 80 answers carry no supporting triples at all. Ten of them are hedges with no claims, for which zero is the honest count. The other three had every claim rejected or certified only through the route that records nothing.

I. A Worked Example

The question "Which university, founded in 1701, did actor James Franco attend?" shows the layer end to end. Its gold answer is the graph's University Yale. Navigation anchors on James Franco and expands his education relations. The plan's first objective, the university founded in 1701, is never satisfied, because the environment holds essentially no date literals. After three evaluator-triggered backtracks, the run stops on the depth budget.

The drafter writes "James Franco attended University Yale, which was founded in 1701." Its entity list is [University Yale], copied from the fact window in the graph's spelling. It decomposes the draft into an attendance claim and a founding claim. Traversed triples join the attendance claim's endpoints, so that claim is supported, and four triples are recorded as its evidence. The founding claim names 1701, which no traversed triple mentions. So it falls to the entailment check, which correctly reports it unsupported. The depth budget is spent, so the router takes give_up. The filter keeps University Yale, since it appears in a supported claim. The rewrite asserts the attendance and hedges the founding date, and the answer scores as a hit.

The instructive part is the counterfactual. Before deterministic filtering, the identical trace ended with [Yale University]. The rewrite call had restated the answer in the model's own spelling

rather than the graph's, and the answer was scored wrong. The claim-level checks had worked perfectly. The layer still emitted an entity string that appears nowhere in the environment, which is the precise thing it exists to prevent.

J. What the Layer Can Get Wrong

The ways the layer can be wrong follow from its design, independently of how often they occur. A claim is *wrongly rejected* when the layer withholds support that the environment would in fact provide. It is *wrongly accepted* when the layer certifies a claim that the environment does not support. Section VII reports which of these mechanisms actually fired.

Five mechanisms lead to wrongful rejection. The most expensive is the composite claim. When a draft asserts a filtered set, such as "these teams were founded before 1996", its claims are about the founding. The membership that makes each team a candidate may never be claimed at all. So each team's only claim is the unverifiable one, the filter drops every team, and a draft with four correct answers becomes a hedge. A claim can also name an endpoint outside the traversal, which skips both structural routes whatever the graph contains. The entailment check's fact window is ranked for relevance to the question rather than to the claim, so a peripheral claim may be judged against facts that never mention it. A failed entailment call records an infrastructure error as a rejection. And the name map keeps the first identifier it sees for a name, so two entities that share a surface form collapse into one.

Wrongful acceptance has more mechanisms, and they reach further. Both structural routes ignore the relation and the direction. So any edge between two entities certifies any claimed relation between them. A claim that X is Y's mother is accepted on an edge recording that X is Y's child. That is the price of free-text relations, and it is the layer's main acceptance risk. On the test sets the verifier accepted $2{,}008$ of the $2{,}110$ claims it decomposed. The number accepted by a relation-blind test lies somewhere in $[39, 2{,}008]$. The lower end is what the log pins down, and the upper end is what the mechanism permits. The mediator hop adds a second looseness. An actor, a film, a character, and a period are all mutually connected through one performance node, so any claim relating two of them is certified. A draft with no claims has no unsupported claim and is routed grounded, which describes 10 of the 80 development questions. An entity that the drafter named but the decomposer never put into a claim is never checked. So the guarantee over emitted entities rests on decomposition recall, which is itself an unchecked model output. A claim accepted structurally is never examined for meaning, since the entailment check sees only what the structural routes could not resolve. And an answer whose every claim is supported can still fail to answer the question, as the New Jersey senators draft does. No claim-level check can catch that, because the defect lies between the claims and the question.

The layer establishes that the entity pairs a draft relates are connected in the environment. On the repair path it also makes the emitted entity list a deterministic function of those verdicts rather than a second generation. It does not establish that the claimed relation is the one the edge records, that an entity absent from every claim was grounded, or that the answer is responsive to the question. Those boundaries are properties of the design rather than defects discovered in it.

V. Experimental Setup

Each research question is answered by a distinct experiment rather than by one table read three ways. RQ1 is answered by the main comparison of five systems on two datasets, scored with Hits@1 and F1, broken down by hop stratum, and tested in pairs. The claim about multi-hop accuracy rests on the per-stratum breakdown, not on the aggregate. RQ2 is answered by three measurements that disagree with one another, and that disagreement is itself the finding. They are the two-tier groundedness metric, the precision columns of the main comparison, and the verifier ablation. RQ3 is answered by four ablations that each remove one component, compared with the unmodified system on the same questions.

A. Pre-Specification

One rule governed everything from the moment the test files were built. Nothing was tuned. No prompt was edited, no configuration value was changed, and no threshold was adjusted after test data was touched. Four decisions were fixed in writing before the measurements they govern. They were the frozen configuration ($\alpha = 0.7$, $\tau = 0.20$, claim verification on), the sample size of 400 questions per dataset, a certification threshold for the retrieval baselines, and a judge-agreement threshold of Cohen's $\kappa \geq 0.7$. The last was missed by 0.0005 and is reported as missed. Nothing was filed with a public registry, so the claim is pre-specification rather than pre-registration.

B. Datasets and Certified Test Samples

WebQSP [9] holds natural questions from search logs, mostly one or two hops from the topic entity, and multi-valued answers are common. ComplexWebQuestions (CWQ) [10] composes WebQSP questions into deeper chains through SPARQL templates, with conjunctions, superlatives, and comparisons. Running five systems over the full test splits of $1{,}628$ and $3{,}531$ questions was projected to exceed the budget. So the evaluation fell back to the pre-specified alternative, a sample of 400 questions per dataset, stratified by hop count at seed 42. In the end the whole benchmark, ablations included, cost $\$11.13$ in API spend.

The stratification variable is the length of the shortest path from any topic entity to any gold answer, capped at four raw hops (Table VI). WebQSP is 64% one-hop, and its stratum of three or more hops holds four questions, too few to interpret. CWQ is mostly two-hop, with a real deep tail of 49 questions. The unreachable stratum is kept. Dropping it would raise every score by the same amount and misdescribe the environment. The ceiling is the share of questions with a reachable gold answer, and no system can score above it on Hits@1. The two benchmarks also differ in their gold answer sets. WebQSP averages 7.15 gold entities per question, with a median of 1.5 and a largest question listing 237. CWQ averages 1.75 with a median of 1.

TABLE VI
THE TWO CERTIFIED TEST SAMPLES BY HOP STRATUM. THE CEILING IS THE SHARE OF THE 400 QUESTIONS WITH A GOLD ANSWER REACHABLE WITHIN FOUR RAW HOPS.

Dataset	n	1 hop	2 hops	3+ hops	Unreach.	Ceiling
WebQSP	400	256	128	4	12	97.0%
CWQ	400	137	211	49	3	99.2%

TABLE VII
GOLD-LABEL ADJUDICATION. THE PASS EMITS ONE ROW PER CONSENSUS ANSWER, SO ROWS AND QUESTIONS ARE DIFFERENT UNITS. EXCLUSIONS ARE THE QUESTIONS JUDGED TO HAVE A WRONG GOLD LABEL OR AN AMBIGUOUS QUESTION.

Dataset	Rows	Flagged	Gold wrong	Ambiguous	Excluded
WebQSP	89	58	12	10	22 (5.5%)
CWQ	59	47	17	2	19 (4.8%)

C. Gold-Label Quality Control

Both benchmarks carry label defects, CWQ more, since its questions and answers are generated by templates. A question whose gold answer is wrong is not a system failure. Counting it as one corrupts both the accuracy tables and the error analysis. The detection principle is consensus against gold. When at least three of the five systems assert the same non-gold answer, the label is worth examining. The voters are the five systems of the main comparison, spanning two knowledge sources and four retrieval paradigms. The ablation conditions would not do, since their agreement would describe one architecture rather than a label. That pass flagged 58 WebQSP and 47 CWQ questions (Table VII). Adjudication confirmed 22 and 19 of them (5.5% and 4.8% of the samples) as a wrong gold label or an underdetermined question. The gap between the flagged and the confirmed counts is not noise, and Section VII-D explains it.

Label errors come in two kinds with opposite evidence signatures, and telling them apart is what makes the adjudication reproducible. A *world-fact error*, where the gold answer contradicts reality, shows *mixed* consensus. The parametric control and the graph-based systems agree against the label, so two different knowledge sources agree. In one specimen, four systems answer Bexar County for the seat of the San Antonio City Council, against a gold answer of Comal County. An *annotation-pipeline error*, where the gold answer contradicts the very graph the labels came from, shows *graph-only* consensus. Asked who had a concert named Crazy Love Tour, the graph-reading systems all return Michael Bublé. The gold answer lists his professions, because the annotation query traversed one relation too far.

The confirmed defects are removed from the failure census. The accuracy tables are not rescored. Rescoring on self-adjudicated labels would replace one contestable ground truth with another, and the defects cut both ways. Recomputing every cell without them moves each system by between $+0.001$ and $+0.020$ Hits@1 and changes no ranking.

D. Backbone

Every system runs on the same frozen backbone, gpt-5.4-mini-2026-03-17, at temperature 0, with reasoning effort set to none and the response cache enabled. A non-reasoning model keeps hidden reasoning tokens out of the cost metric. Temperature zero on a hosted API is greedy rather than deterministic, and a sequential agent amplifies a single divergent token. In a qualification probe, the frozen model repeated its full decision sequence on 8 of 12 probes. The probe compared it with gpt-4.1-mini-2025-04-14, and both models carried a persistent ungrounded answer. One asserted the United States dollar for a question about the Dominican peso in two of four runs, and the other named the wrong Beckham child in all four. Each also carried intermittent ones, where the same question on the same graph moved between a grounded correct answer and an ungrounded wrong one. Choosing a backbone does not remove this class of error. It only moves it around, which is the premise of this work. So reproducibility rests on the cache. Every table comes from one recorded run per condition, which replays exactly from the cache. Run-to-run variance is not measured, and Section VIII-D carries it as a limitation.

E. Baselines

Four baselines span the space between no retrieval and agentic navigation.

The **no-retrieval** control makes a single call with no facts and may return an empty answer. WebQSP and CWQ predate current models and may sit in their training data. So this control measures the floor that every retrieval result is read against.

Vector-RAG verbalises every triple as a sentence, embeds it with MiniLM [13], and answers from the top 30 sentences in one call. That is dense retrieval [12] applied to a graph. A fact stored through a mediator has no single triple that expresses it, so no retrieved sentence can express it either.

Static GraphRAG is the comparison with structure but without navigation [4]. It expands a one-logical-hop neighbourhood around the topic entities, reranks it by similarity to the question, and answers from the top 30 facts. It shares Vector-RAG's prompt, so the difference between the two isolates the retrieval strategy. Its expansion keeps at most 100 edges per topic entity in no stated order, and at least one topic entity is truncated on 72.5% of questions.

Think-on-Graph is the agentic baseline [5], reimplemented on this environment's tools. It uses beam width 3, depth 3, model-based pruning of relations and entities, and a sufficiency check at each depth. Holding the graph, model, and budget constant makes it a stronger comparison than citing published numbers. It also makes its numbers incomparable with the published ones. Two asymmetries are stated in advance. The baseline stops its search one call before the shared limit and reserves that call for its answer, where AGR makes no such reservation. And it keeps the first 40 relations per entity and 20 neighbours per relation, its own algorithm's pruning widths, against AGR's 300 and 200. That cut binds on 31.6% of the entities it expands, against 1 of AGR's 3,097.

Every system shares the backbone, the environment and its relation blocklist, the entity resolver, the question files, the

25-call budget, and the output schema. Two things are not shared. The baselines get a plain answering prompt, while AGR's drafting prompt carries the provisions of Section IV-H. And the candidate widths differ, as just stated. Each retrieval baseline was certified on development data before any test question ran. The pre-specified failure was hedging while the gold answer sat in the baseline's own retrieved facts, allowed on fewer than 15% of hedges. Vector-RAG measured 0% on both datasets, and Static GraphRAG 10% and 14%.

F. Metrics

Both accuracy metrics are computed over entity sets after Unicode NFKC normalisation, case folding, and whitespace stripping, with no fuzzy matching. *Hits@1* is 1 if any predicted entity matches any gold entity, which is the convention in this literature. *F1* is computed per question between the predicted set A and the gold set A^* and averaged over questions, with $P = |A \cap A^*|/|A|$, $R = |A \cap A^*|/|A^*|$, and $F_1 = 2PR/(P + R)$. Hits@1 rewards a system that names one correct entity among several wrong ones. Precision charges for the rest. Strict matching does not manufacture failures. Only 1 of AGR's 65 wrong answers on WebQSP and 6 of its 99 on CWQ are surface-form near misses.

A *hedge* is an empty answer set. It scores zero on both metrics, and the hedge rate gets its own column, so the trade between abstention and accuracy stays visible. *Assertion precision* is the share of the questions a system commits on that it answers correctly.

Groundedness is measured in two tiers, and the word *tier* refers to this metric and nothing else. **Tier 1** is structural and deterministic. An asserted entity is grounded if and only if it exists as a node and lies within four hops of a topic entity. It applies to every asserted entity in all $4{,}000$ records. **Tier 2** asks whether the entity is supported *as the answer*. For 60 asserted answers per system per dataset, a model judges whether any of the three shortest graph paths between topic and answer expresses the relationship the question asks about. Mere connectedness is not support. The judge sees no system identity, gold answer, or score. The first author labelled 100 judged items blind, without opening the judge's verdicts or the answer key until the labels were saved. Observed agreement was 85%, with near-symmetric marginals (54 items supported by the human against 51 by the judge) and near-symmetric disagreement (9 against 6). So neither rater is systematically the more lenient. Cohen's κ [38] corrects that agreement for chance,

$$\kappa = \frac{p_o - p_e}{1 - p_e} = \frac{0.8500 - 0.5008}{1 - 0.5008} = 0.6995, \qquad (2)$$

where p_o is the observed agreement and p_e the agreement expected by chance. That is substantial agreement on the conventional scale [39]. But it falls short of the pre-specified 0.7 by 0.0005, and it is reported as short. Re-judging a fresh sample until the bar was cleared was available and was not taken, since re-running until a threshold is met is exactly what pre-specification exists to prevent. So no conclusion rests on Tier 2 alone. The judge also runs on the same backbone as the systems it judges.

Tokens and calls are counted by the shared budget meter and are exact. Latency is computed over cold-cache records only. Confidence intervals are 95% percentile bootstrap intervals [40] from $10{,}000$ resamples. They are read as a sense of scale for a difference, not as a hypothesis test, and they make no distributional assumption about per-question F1. Paired comparisons use the exact form of McNemar's test [41] on per-question correctness. The test looks only at the discordant pairs, the questions that one system answers correctly and the other does not. Pairing matters here, because every system runs on the same questions, and an unpaired test would throw that pairing away.

G. Ablations and Development Tuning

Four conditions each remove one mechanism. **No-planner** installs the whole question as the only sub-objective. **No-backtrack** sets the backtrack budget to zero. **No-verifier** emits the draft without checking any claim, with the drafting call unchanged. **Embedding-only** sets $\alpha = 1$, which removes the model term from the scorer. The ablations run on stratified halves of 200 WebQSP and 198 CWQ questions. The reference row is the unmodified system restricted to the same questions. Accuracy on 400 questions and on 198 are different numbers for the same system, so comparing against the full score would be invalid. The planner flag was the only code change. It was certified as a no-op on the frozen path, since re-running an answered question served every model call from the cache.

The two free parameters were tuned on 80 development questions, disjoint from both test samples. A joint sweep over $\alpha \in \{0.3, 0.5, 0.7, 1.0\}$ and $\tau \in \{0, 0.20\}$ peaked at $\alpha = 0.7$, with 64 of 80 questions correct against 61, 62, and 59 at the other three weights (Table VIII). The curve is not clean. The wrong-answer column peaks at $\alpha = 0.5$, where the system commits more often. The threshold τ barely mattered at the values swept, and it was frozen at 0.20 anyway. That is stated plainly rather than dressed up as a tuned choice. Two further conditions switched claim checking off. At the frozen weights the draft-only condition scored 66 hits to 64. After the answer-entity filter of Section IV-H was corrected, still on development data, the frozen condition scored 65 hits, 11 hedges, and 4 wrong answers. So on this set the layer cost one correct answer and removed no wrong one. The draft-only condition was never re-run after the correction, so this is the closest available reading rather than a matched pair. Section VI-G finds the same thing at test scale.

VI. Results

A. Main Results

On WebQSP, AGR answers 302 of 400 questions correctly, a Hits@1 of 0.755 against a ceiling of 97.0% (Table IX). That is 9.5 points above Think-on-Graph and 30.2 above the parametric floor. The floor matters. The backbone answers 45.3% of these questions from memory alone. So any claim about what retrieval contributes starts from 0.453, not from zero.

F1 widens the margin that Hits@1 understates. Against Think-on-Graph the Hits@1 gap is 9.5 points and the F1 gap

TABLE VIII
THE DEVELOPMENT SWEEP, $n = 80$. HITS, HEDGES, AND WRONG ANSWERS PARTITION THE SET. THE LAST TWO ROWS SWITCH CLAIM CHECKING OFF. ALL ROWS PREDATE THE CORRECTION TO THE ANSWER-ENTITY FILTER.

α	τ	Verify	Hits	Hedge	Wrong	F1	Tokens	Calls
0.3	0.00	yes	61	12	7	0.728	4,999	7.4
0.3	0.20	yes	61	12	7	0.728	5,030	7.5
0.5	0.00	yes	62	10	8	0.743	4,874	7.2
0.5	0.20	yes	62	10	8	0.743	4,874	7.2
0.7	0.00	yes	64	11	5	0.769	4,925	7.3
0.7	**0.20**	**yes**	**64**	11	5	0.769	4,925	7.3
1.0	0.00	yes	59	15	6	0.713	3,408	4.9
1.0	0.20	yes	59	16	5	0.713	3,400	4.9
0.5	0.20	no	64	9	7	0.770	4,807	7.1
0.7	0.20	no	66	10	4	0.797	4,858	7.2

16.5. The precision column, at 0.697 against 0.571, shows where the extra margin comes from. WebQSP questions carry seven gold answers on average. So a system that asserts a long list collects Hits@1 credit for the one entity that matches and pays nothing for the rest. Precision charges for the rest. Together the two columns say that AGR answers more questions and asserts fewer falsehoods. It does so at an 8.2% hedge rate, the lowest of the five systems, so its precision is not bought with abstention.

Raw hits mislead on the retrieval baselines. GraphRAG scores 0.340, below the no-retrieval control's 0.453. But the control asserts on 351 questions and is wrong on 170, an assertion precision of 51.6%. GraphRAG asserts on 177 and is wrong on 41, for 76.8%. The retrieval baselines are told to answer only from the facts given, so they abstain when retrieval misses. The control guesses from memory, and on a memorised benchmark guessing pays. A grounded system that scores below an ungrounded one on raw hits, while committing four times fewer falsehoods, is the contamination the control was built to expose.

On CWQ, AGR answers 209 of 400, a Hits@1 of 0.522 against a ceiling of 99.2%. That is 8.7 points above Think-on-Graph and 21.5 above the floor. Both static baselines now fall below the parametric control. Vector-RAG drops from 0.568 to 0.203 and GraphRAG from 0.340 to 0.205, while no-retrieval falls only from 0.453 to 0.307. Vector-RAG's context is one verbalised triple, which cannot hold a chain at any radius. GraphRAG's one-hop radius confounds its result, so the claim rests on Vector-RAG. The two agentic systems commit at nearly the same rate, on 308 and 310 questions. From those commitments AGR is right 209 times and wrong 99, against 174 and 136 for Think-on-Graph. That is an assertion precision of 67.9% against 56.1%.

All eight paired comparisons reject (Table X). That includes the two against Think-on-Graph, the only baseline that navigates the same graph with the same model under the same budget. On CWQ, AGR uniquely solves 86 questions that Think-on-Graph misses, against 51 the other way.

B. Where the Margin over the Agentic Baseline Comes From

The shared call cap binds on Think-on-Graph far more often than on AGR, because beam search costs width times depth. So the same budget is not the same freedom to search. Splitting the questions on whether the cap actually cut Think-on-Graph off separates search quality from affordability (Table XI).

On the questions it is allowed to finish, Think-on-Graph is ahead on both datasets. It scores 0.852 against AGR's 0.788 on WebQSP and 0.629 against 0.607 on CWQ. The entire aggregate margin comes from the 29.2% and 44.0% of questions where the cap truncates it. There its accuracy falls to 0.197 and 0.188, while AGR's holds at 0.675 and 0.415.

So the finding is not that guided best-first search resolves entities better than model-pruned beam search. On this evidence it does not. The finding is that beam search cannot afford to finish under a budget a deployed system would impose, and guided search can. Width times depth multiplies, and every question pays it. A plan plus a scored frontier does not multiply. The budget was fixed in advance because cost is a property of the method, not a nuisance to control away.

Two qualifications keep the claim honest. The cap of 25 calls is itself a choice, and a larger one would move the split. And Think-on-Graph prunes from narrower candidate sets than AGR (Section V-E). That cannot explain the clipping, since a thinner candidate set makes each step cheaper. But it does make the unclipped scores of 0.852 and 0.629 a lower bound on what the baseline's search could resolve with AGR's widths.

The breadth of AGR's answers is checked rather than assumed. On answered WebQSP questions AGR names 3.37 entities against Think-on-Graph's 2.56, and a metric that scores a hit on any match rewards naming more. But AGR's entity precision on the same questions is higher too, at 0.760 against 0.703. On CWQ the figures are 1.54 against 1.40 entities and 0.629 against 0.520 precision. If AGR bought breadth without accuracy, its precision would fall as its entity count rose. It does not fall.

C. Positioning against Published RoG Results

The environment is built from the subgraphs that RoG [7] publishes, so the two systems invite a direct comparison. Table XII gives it, and it does not favour this work. RoG leads on all four figures, and the margin is not sampling noise. AGR's 95% intervals on Hits@1 are $[71.3, 79.7]$ and $[47.5, 57.0]$, and RoG's 85.7 and 62.6 fall outside both.

Four differences bound what the comparison establishes. The first is decisive. RoG is trained on these benchmarks, with its backbone fine-tuned on training splits that hold 2,826 and 27,639 questions in this distribution. AGR performs no training of any kind. RoG also searches each question's own subgraph, where AGR searches the union of them all. The backbones differ. And RoG reports over the full test splits rather than over samples of them. A fifth difference, the scorer, is measured and small. RoG's released scorer counts a gold answer found anywhere inside the predicted text. Scored with its functions, no figure of AGR's moves by as much as a point (Table XII). So the comparison places AGR. A training-free system reaches 75.5 and 52.2 against a supervised system's 85.7 and 62.6, on the same data, with no in-domain examples. The contribution of this work lies outside the accuracy column.

TABLE IX
MAIN RESULTS, $n = 400$ PER DATASET. BRACKETS ARE 95% BOOTSTRAP INTERVALS ON HITS@1. P AND R ARE MEAN PER-QUESTION PRECISION AND RECALL, AND HEDGE IS THE SHARE OF QUESTIONS WITH AN EMPTY ANSWER. THE HITS@1 CEILINGS ARE 97.0% AND 99.2%.

	WebQSP					CWQ				
System	Hits@1	F1	P	R	Hedge	Hits@1	F1	P	R	Hedge
No-retrieval	0.453 [0.403, 0.500]	0.305	0.380	0.303	12.2%	0.307 [0.263, 0.352]	0.279	0.285	0.282	38.0%
Vector-RAG	0.568 [0.517, 0.618]	0.432	0.469	0.460	27.0%	0.203 [0.163, 0.242]	0.168	0.171	0.189	48.2%
GraphRAG	0.340 [0.292, 0.388]	0.262	0.279	0.284	55.8%	0.205 [0.168, 0.245]	0.183	0.187	0.194	60.8%
Think-on-Graph	0.660 [0.613, 0.705]	0.477	0.571	0.480	18.8%	0.435 [0.388, 0.482]	0.378	0.403	0.384	**22.5%**
AGR	**0.755** [0.713, 0.797]	**0.642**	**0.697**	**0.653**	**8.2%**	**0.522** [0.475, 0.570]	**0.469**	**0.484**	**0.481**	23.0%

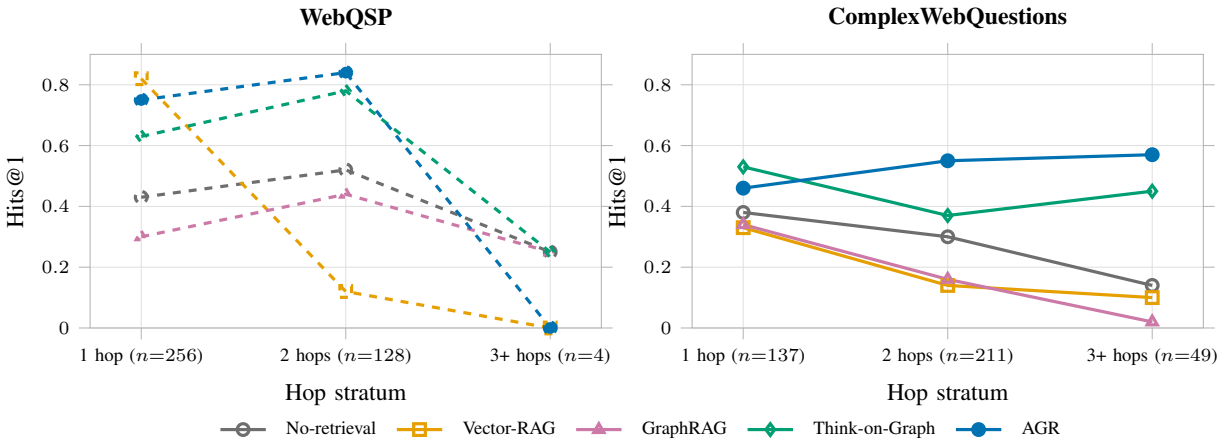


Fig. 3. Hits@1 across hop strata. On ComplexWebQuestions, AGR is the only system whose accuracy rises with hop count. WebQSP is drawn dashed because its stratum of three or more hops holds four questions and cannot carry a trend. The unreachable stratum is omitted, since it measures assertion rather than accuracy.

TABLE X
PAIRED MCNEMAR TESTS OF AGR AGAINST EACH BASELINE ON THE SAME 400 QUESTIONS. "AGR ONLY" COUNTS THE QUESTIONS THAT ONLY AGR ANSWERS CORRECTLY, AND "BASE ONLY" COUNTS THE REVERSE.

Data	Baseline	AGR only	Base only	p
WebQSP	No-retrieval	152	31	2.3×10^{-20}
WebQSP	Vector-RAG	117	42	2.2×10^{-9}
WebQSP	GraphRAG	186	20	6.5×10^{-35}
WebQSP	Think-on-Graph	76	38	4.8×10^{-4}
CWQ	No-retrieval	121	35	2.7×10^{-12}
CWQ	Vector-RAG	151	23	2.8×10^{-24}
CWQ	GraphRAG	148	21	1.0×10^{-24}
CWQ	Think-on-Graph	86	51	3.5×10^{-3}

TABLE XI
THE COMPARISON AGAINST THINK-ON-GRAPH, SPLIT ON WHETHER THE SHARED 25-CALL CAP CUT IT SHORT. THE COMBINED ROWS REPRODUCE THE HITS@1 COLUMN OF TABLE IX.

Data	Subset	n	ToG	AGR	Δ
WebQSP	ToG ran to completion	283	0.852	0.788	−0.064
	ToG clipped by the cap	117	0.197	0.675	+0.478
	Combined	400	0.660	0.755	+0.095
CWQ	ToG ran to completion	224	0.629	0.607	−0.022
	ToG clipped by the cap	176	0.188	0.415	+0.227
	Combined	400	0.435	0.522	+0.087

TABLE XII
AGR AGAINST ROG'S PUBLISHED FIGURES, IN POINTS. ROG'S ROW IS MEASURED OVER THE FULL TEST SPLITS AND AGR'S OVER THE CERTIFIED SAMPLES. THE LAST ROW SCORES AGR'S ANSWERS WITH ROG'S RELEASED SCORER.

	WebQSP		CWQ	
System	Hits@1	F1	Hits@1	F1
RoG (published)	85.7	70.8	62.6	56.2
AGR	75.5	64.2	52.2	46.9
AGR, scored by RoG's code	76.0	64.9	52.2	47.2

TABLE XIII
HITS@1 AND F1 PER HOP STRATUM, WRITTEN AS HITS@1/F1. STRATUM SIZES ARE THOSE OF TABLE VI. WEBQSP'S STRATUM OF THREE OR MORE HOPS HOLDS FOUR QUESTIONS AND IS NOT INTERPRETED.

Data	System	1 hop	2 hops	3+ hops	Unreach.
WebQSP	No-retrieval	0.43/0.29	0.52/0.36	0.25/0.00	0.17/0.17
WebQSP	Vector-RAG	0.82/0.63	0.12/0.07	0.00/0.00	0.08/0.08
WebQSP	GraphRAG	0.30/0.23	0.44/0.35	0.25/0.07	0.08/0.08
WebQSP	Think-on-Graph	0.63/0.45	0.78/0.58	0.25/0.07	0.17/0.17
WebQSP	AGR	0.75/0.64	**0.84/0.73**	0.00/0.00	0.08/0.08
CWQ	No-retrieval	0.38/0.35	0.30/0.27	0.14/0.14	0.00/0.00
CWQ	Vector-RAG	0.33/0.27	0.14/0.11	0.10/0.09	0.67/0.67
CWQ	GraphRAG	0.34/0.30	0.16/0.14	0.02/0.02	0.33/0.33
CWQ	Think-on-Graph	**0.53**/0.48	0.37/0.32	0.45/0.32	0.33/0.33
CWQ	AGR	0.46/0.42	**0.55/0.49**	**0.57/0.50**	0.33/0.33

D. Accuracy by Hop Count

Fig. 3 and Table XIII answer RQ1 with a shape rather than a difference of means. On CWQ, AGR's accuracy rises with hop

count. Its Hits@1 goes $0.46 \rightarrow 0.55 \rightarrow 0.57$ across one, two, and three or more hops, and its F1 goes $0.42 \rightarrow 0.49 \rightarrow 0.50$. It is the only system on that dataset that ends above where it started. Three of the others decay monotonically. No-retrieval goes $0.38 \rightarrow 0.30 \rightarrow 0.14$, Vector-RAG $0.33 \rightarrow 0.14 \rightarrow 0.10$, and GraphRAG $0.34 \rightarrow 0.16 \rightarrow 0.02$. Think-on-Graph falls from 0.53 to 0.37 and partly recovers to 0.45, still below its own one-hop score. A system that gets better as the required chain gets longer is not merely ahead on average. Its profile is the only one that fits a system which actually traverses the chain.

Two cells cut the other way, and they are reported for that reason. Think-on-Graph beats AGR on CWQ's one-hop stratum, at 0.53 against 0.46 over 137 questions. The ablation of Section VI-G identifies the mechanism. Decomposition costs accuracy on questions that do not need decomposing. And Vector-RAG scores 0.82 on WebQSP's one-hop stratum, the best single-hop number anywhere, but 0.12 at two hops. When one triple suffices, dense retrieval over verbalised triples is excellent. When a chain is required, it has nothing to offer.

GraphRAG's WebQSP profile is inverted, at 0.30 on one hop against 0.44 on two. Part of that is hub noise. A popular entity's neighbourhood drowns the answer, or the unordered sample of its edges leaves the answer out. The rest marks the radius. WebQSP's two-hop stratum is largely mediator paths, which a one-hop expansion reaches. CWQ's two-hop stratum is genuine composition, where GraphRAG scores 0.16. So its row measures what its radius covers, not how far static retrieval can see.

The unreachable stratum is a hallucination probe rather than an accuracy measurement. For these questions no path to gold exists within the hop bound. So every hit there is an assertion the environment cannot support, one that happened to match the label. Vector-RAG scores 0.67 on CWQ's three unreachable questions, and no-retrieval 0.17 on WebQSP's twelve. The counts are tiny and carry no weight. But the direction is the one Section VI-E measures at scale.

E. Groundedness

Table XIV reports both tiers. The structural differential is emphatic. The parametric control asserts 661 entities on WebQSP, of which 179 (27.1%) have no basis in the environment. Both navigating systems sit at exactly zero, over $1{,}235$ entities for AGR and 832 for Think-on-Graph. Graph-navigated answering removes structural hallucination outright.

But Think-on-Graph reaches zero as well, and that changes the attribution. Structural groundedness is a property of the navigation paradigm. Any system that copies its answer entities out of triples it traversed inherits it. It is not a property of the verification layer. The two navigating systems differ at the level of semantic support. Think-on-Graph's precision of 0.571 and 0.403, against AGR's 0.697 and 0.484, shows it asserting more entities that exist and connect to the topic but are not the answer. Vector-RAG's few ungrounded entities are mostly retrieval errors rather than fabrications. Its index is global, so retrieval can surface a similarly named subject elsewhere in the graph. And the parametric control's CWQ rate of 12.4% is less than half its WebQSP rate. That is not because it fabricates less on harder questions. It hedges more, at 38.0% against 12.2%, and abstention pushes the measured rate down.

TABLE XIV
TWO-TIER GROUNDEDNESS. TIER 1 COUNTS ASSERTED ENTITIES THAT ARE ABSENT FROM THE GRAPH OR MORE THAN FOUR HOPS FROM EVERY TOPIC ENTITY. TIER 2 IS THE JUDGED SHARE OF 60 SAMPLED ANSWERS PER CELL THAT ARE SUPPORTED AS THE ANSWER.

Data	System	Tier 1 ungrounded	Tier 2 supported
WebQSP	No-retrieval	179 of 661 (27.1%)	63.3%
WebQSP	Vector-RAG	38 of 1,040 (3.7%)	61.7%
WebQSP	GraphRAG	6 of 800 (0.8%)	55.0%
WebQSP	Think-on-Graph	0 of 832 (0.0%)	61.7%
WebQSP	AGR	0 of 1,235 (0.0%)	66.7%
CWQ	No-retrieval	42 of 340 (12.4%)	36.7%
CWQ	Vector-RAG	7 of 289 (2.4%)	50.0%
CWQ	GraphRAG	2 of 224 (0.9%)	36.7%
CWQ	Think-on-Graph	0 of 433 (0.0%)	43.3%
CWQ	AGR	0 of 474 (0.0%)	48.3%

TABLE XV
COST PER QUESTION. SECONDS USE COLD-CACHE RECORDS ONLY. CLIPPED IS THE SHARE OF QUESTIONS ON WHICH THE SHARED 25-CALL BUDGET RAN OUT MID-SEARCH.

Data	System	Tokens	Calls	Seconds	Clipped
WebQSP	No-retrieval	113	1.0	1.2	—
WebQSP	Vector-RAG	527	1.0	1.4	—
WebQSP	GraphRAG	531	1.0	2.1	—
WebQSP	Think-on-Graph	3,615	12.8	12.9	29.2%
WebQSP	AGR	4,511	6.2	11.2	0.0%
CWQ	No-retrieval	118	1.0	1.0	—
CWQ	Vector-RAG	535	1.0	1.4	—
CWQ	GraphRAG	470	1.0	2.2	—
CWQ	Think-on-Graph	5,572	18.2	15.2	44.0%
CWQ	AGR	6,818	8.9	17.2	0.0%

Tier 2 places every system in a band from 36.7% to 66.7%. AGR is highest on WebQSP and second on CWQ, 5.0 points ahead of Think-on-Graph on both. Neither gap is resolvable at this sample size. With 60 judgements per cell, the 95% interval has a half-width of about 12 to 13 points. What the tier does establish is the level. Zero structural hallucination coexists with a third to a half of assertions failing a strict test of semantic support. That is a statement about the paradigm, not about any one system.

F. Efficiency

Agency costs roughly ten times what one-shot retrieval costs (Table XV). AGR takes $4{,}511$ and $6{,}818$ tokens per question against about 500, and 11 to 17 seconds against one or two. That is the price of RQ1's accuracy, and it is stated as a price.

The informative comparison is with the other agentic system. AGR uses half the calls, 6.2 against 12.8 on WebQSP and 8.9 against 18.2 on CWQ. It spends more tokens per call, because each call carries a plan, a scored candidate set, and a fact window. Under the shared budget, Think-on-Graph exhausts its calls on 29.2% and 44.0% of questions, while AGR never reaches the call cap. AGR's own budgets do bind. The depth cap refuses a deeper expansion on 25.6% of test questions, against 6.2% for backtracks and 1.9% for repair.

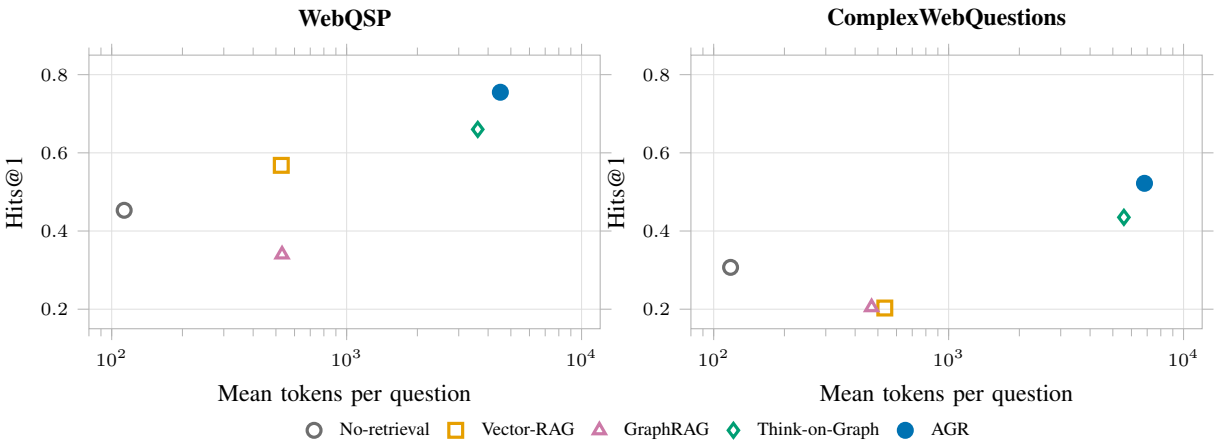


Fig. 4. Accuracy against token cost, one point per system, on a logarithmic token axis. Think-on-Graph sits below and to the left of AGR, so it is a frontier point on tokens, not a dominated one. The domination claim holds on calls, which are not plotted and are given in Table XV.

The cost frontier depends on which cost is charged (Fig. 4). On tokens, Think-on-Graph is not dominated. It buys its lower accuracy at a lower token price on both datasets. On calls it is dominated, with lower Hits@1 at about twice the calls. Calls are what the budget meters, and it is the call budget that clips the baseline. Vector-RAG on WebQSP is the cheap corner, the right choice for shallow questions and useless for deep ones.

G. Ablations

Table XVI gives the four conditions. One reaches significance, and it points the opposite way from what the design intended.

The planner. On WebQSP, removing the planner improves every axis at once. Hits@1 goes from 0.755 to 0.830 and F1 from 0.659 to 0.742, while tokens fall from 4,562 to 3,159 and calls from 6.2 to 4.0. McNemar gives $p = 0.006$ on 6 against 21 discordant pairs. Without the planner, the one-hop stratum rises from 0.75 to 0.85 and the two-hop stratum from 0.84 to 0.89. On CWQ the sign reverses, at $0.495 \rightarrow 0.434$ on 27 against 15 discordant pairs ($p = 0.088$), short of significance. There the one-hop stratum still improves ($0.49 \rightarrow 0.54$), while the two-hop stratum collapses ($0.48 \rightarrow 0.34$). That is the pattern of decomposition that helps genuine composition and hurts questions that do not need it. So the planner's value depends on question structure. Applying it to every question regardless is the defect.

Backtracking. Hits@1 moves by -0.010 and -0.005, with $p = 0.73$ and 1.00. The mechanism is not idle. The full runs record 63 backtracks on WebQSP and 199 on CWQ. But this experiment cannot measure their accuracy consequence. We report no detectable effect at $n \approx 200$, not a confirmed null.

Verification. This is the cleanest null in the table. Across 398 paired questions, removing claim verification changed the correctness of exactly two. Hits@1 moves by $+0.005$ and -0.005, and precision by $+0.001$ and -0.010. The design predicts as much. On 77 of 80 development questions the layer certified the draft and left it alone, and a mechanism that mostly passes cannot move an accuracy metric. Claim verification does not measurably change how often AGR is right. It changes how often AGR is right for a reason it can show.

Embedding-only scoring. Removing the model term costs 0.015 and 0.020 Hits@1. The effect is negative on both datasets and short of significance ($p = 0.66$ and 0.48). The condition is also cheaper on both, so the model term buys accuracy with tokens as designed. The development sweep, where $\alpha = 1$ lost five hits on 80 questions, remains the main evidence for $\alpha = 0.7$. The condition also carries the most backtracks, 57 and 143 against the reference's 38 and 108. Part of that rise is an implementation artefact. With the model term disabled, the signal compared against τ is taken over fewer candidates, so it falls after a ban where the reference's holds steady. So the backtrack count is corroborating, not load-bearing.

One condition of four reaching significance is the correct outcome to report. No correction for multiple comparisons was pre-specified, so the p-values are read as descriptive. Applied after the fact, and labelled as such, a Bonferroni correction changes no claim. The planner's $p = 0.00592$ clears $0.05/8 = 0.00625$ within the eight ablation tests, and the eight baseline comparisons clear the same bar, the largest of them at $p = 0.0035$. The power limit is arithmetic. At $n \approx 200$ and accuracy between 0.5 and 0.75, a component that changes five questions in a hundred produces on the order of ten discordant pairs. That is not enough to reject. The planner cleared the bar because its effect is large, at 21 against 6. So three conditions are reported as having no detectable effect at this sample size, which is not the same as no effect.

VII. Error Analysis

Section VI measured how often each system fails. This section reads the failures and names the mechanism of each one. The census is a population, not a sample. Every question the full system did not answer correctly was read and labelled by hand, after the confirmed benchmark defects were removed, for 256 failures in all. The reading surfaced 16 more benchmark defects inside the census. They are filed as gold noise or ambiguous questions, so those two rows of Table XVII count the benchmark's failures rather than the system's. Each failure

TABLE XVI
ABLATION CONDITIONS ON STRATIFIED HALVES ($n = 200$ WEBQSP, $n = 198$ CWQ). THE REFERENCE ROW IS THE FROZEN SYSTEM RESTRICTED TO THE SAME QUESTIONS. P IS MEAN PER-QUESTION PRECISION, AND p IS THE EXACT MCNEMAR TEST AGAINST THE REFERENCE ON PER-QUESTION CORRECTNESS.

	WebQSP							CWQ						
Condition	Hits@1	F1	P	Hedge	Tokens	Calls	p	Hits@1	F1	P	Hedge	Tokens	Calls	p
Reference	0.755	0.659	0.710	8.5%	4,562	6.2	—	0.495	0.445	0.461	23.2%	7,105	9.1	—
No-planner	**0.830**	**0.742**	0.788	5.5%	3,159	4.0	0.006	0.434	0.398	0.407	26.8%	5,617	6.7	0.088
No-backtrack	0.745	0.646	0.704	9.5%	4,261	5.8	0.73	0.490	0.445	0.456	28.3%	5,792	7.4	1.00
No-verifier	0.760	0.663	0.711	8.0%	4,460	6.1	1.00	0.490	0.436	0.451	20.2%	6,704	8.7	1.00
Embedding-only	0.740	0.643	0.701	13.5%	3,413	4.4	0.66	0.475	0.437	0.452	28.3%	4,925	5.9	0.48

TABLE XVII
THE FAILURE CENSUS OF AGR, $n = 256$. EACH FAILURE IS LABELLED AT THE POINT WHERE ITS TRAJECTORY FIRST LEFT THE CORRECT PATH. WRONG ANSWERS AND HEDGES ARE KEPT SEPARATE.

	WebQSP		CWQ		
Category	Wrong	Hedge	Wrong	Hedge	Total
Relation selection	19	6	20	19	64
Composite claim	1	—	23	23	47
Knowledge-graph gap	8	4	7	24	43
Decomposition error	15	11	10	2	38
Answer selection	1	1	8	5	15
Echo attractor	5	2	4	2	13
Ambiguous question	1	1	3	5	10
Premature termination	1	4	1	2	8
Gold noise	1	—	2	3	6
Other	—	1	—	5	6
Verifier, wrongly rejected	—	3	1	1	5
Verifier, wrongly accepted	—	—	—	1	1
Total	52	33	79	92	256

packet holds the question, the gold answer, the answer given, the plan, the explored relations with their scores, the backtrack reasons, and the verifier outcome. The protocol was fixed before reading began. For each packet, find where the trajectory first left the correct path, and label that point rather than the last visible symptom. Without that rule, a run that mis-decomposes, then explores the wrong relation, then hedges would be counted three times.

A. The Shape of the Failures

Table XVII gives the histogram. Relation selection sits at or near the top on both datasets, with 25 of WebQSP's 85 failures and 39 of CWQ's 171. Picking the wrong edge is the base rate of graph navigation. What distinguishes the datasets is a pair of categories that are nearly absent from one and dominant in the other. Composite claims account for 1 failure on WebQSP and 46 on CWQ, and knowledge-graph gaps for 12 and 31. Decomposition errors run the other way, at 26 against 12.

Both asymmetries follow from how the datasets were built. WebQSP is 64% one-hop, so it presents the planner with exactly the questions that do not need decomposing. That mismatch is the significant ablation result of Section VI-G, appearing here as a category count. CWQ's templates invoke dates, numbers, and ordinals constantly, and the environment can express none of them (Section III-D). When those gaps make a CWQ question unanswerable, AGR more often declines than errs, with 24 hedges against 7 wrong answers in that category.

B. Failures inside the Agent

The entity-extraction bug. Nine decomposition errors share one mechanism, and they are the most actionable finding in the census. Asked "who was stephen r covey", the system drafts "Stephen Covey was a Writer, Motivational speaker, Author, Consultant, Professor, and Manager". All six gold values are there. But the scored entity list comes back as the sentence's grammatical subject, [Stephen Covey]. Navigation and verification both succeeded. The step that turns answer text into an entity list failed. So the measured accuracy on this question shape is a floor.

Right candidate, wrong draft. In 15 answer-selection failures the exact gold value entered the resolved candidates, and the drafter chose something else. One was first filed as a verifier error. The draft read "Marlon Brando also played in *Joy*", and the verifier rejected it. Brando was not in *Joy*, but De Niro was, and the evaluator had resolved him. The verifier was correct, and the case is the clearest example in the census of the layer doing its job. A related family of six shares a mechanism nothing predicted. The evaluator resolves the gold value, the verifier returns grounded, and the draft hedges anyway. One answer states that Spanish was spoken in Spain and then that the first language spoken there could not be determined.

Context-stripping. Three decomposition failures show a mechanism that decomposition's own design cannot see. Splitting a question into steps removes a qualifier that appears only in a later clause. So an early sub-objective resolves the wrong sense of an ambiguous term before the disambiguating information arrives. The strongest specimen asks for the 33rd president who "was the nation's leader during WW2". That is one entity under two descriptions. The planner emits two unlinked sub-objectives, as though the descriptions were a chain. With the first one active, the anchor World War II scores at the noise floor, and the war constraint is never used. The run drafts Woodrow Wilson, the 28th president. The undecomposed run answers Harry S. Truman correctly in three calls.

Over-decomposition. The more expected decomposition failures appear too. In one, the plan splits "what type of religion did massachusetts have" into a step that finds Massachusetts and a step that finds its religion type. The run burns 14 of its 16 calls on three evaluator backtracks and hedges, while the undecomposed run finds the same fact in three calls. In another, the planner rewrites "where is ancient phoenician" as the single step "find ancient Phoenician". The rewrite drops

the interrogative, the word that says what is being asked. So the explorer re-identifies the topic instead of locating it, and the run answers "Phoenicia", grounded and useless. With the ask removed, the nearest well-grounded entity to a topic is the topic itself.

Systemic relation gaps. Relation selection is the largest category, at 64 failures. Either a wrong relation was explored, or the right one was explored and abandoned. Its interest lies in cases that show the same gap from several angles. Three questions on CWQ share a stem and reach Beyoncé by three different routes, through a documentary, an actor, and a composer. All three resolve her correctly. None ever tries people.person.children, the single relation that answers each of them, and all three hedge. Three independent entry points hitting the same dead end show that the gap is systemic. It is a property of the scorer's ranking over that entity's relations, not of one question's phrasing.

Abandoned relations. Five premature terminations share one pattern across unrelated domains. The correct relation is explored with a solid score, as high as 0.731, and the evaluator backtracks away from it and never returns. That reads as a policy defect, not a reasoning gap.

Verifier errors. The two error polarities are very unevenly evidenced, and the imbalance comes from the instrumentation. Five failures came from a wrongly rejected claim, all by one mechanism. Each claim was true, but no single edge stated it in the phrasing the drafter used, and the retry lost the answer. Three of the five had already resolved the exact gold answer before the rejection forced the retry. The verifier ran 828 times over the 800 test questions, since a repaired question is verified again. Its first verdict was unsupported on 52 questions. After repairs, 761 questions were answered under a grounded verdict and 39 under an unsupported one. So roughly one rejected question in ten cost a correct answer, and even that is a lower bound. Wrongful acceptance has no logged population at all, because accepted claims are not persisted. One specimen surfaced by hand. A claim connecting Seth MacFarlane to *ThunderCats* was accepted, and a direct Cypher query confirms that no such edge exists. A second case shows why acceptance is hard to judge. Asked when President Wilson was in office, the system answered with inauguration events whose edges are real, since the environment holds no dates. The structural check passed correctly while the answer is wrong. A check of whether a claim is *in* the graph cannot check whether it is *the answer*.

C. Knowledge-Graph Gaps

Knowledge-graph gaps are the third-largest category, at 43 failures, and CWQ's largest hedge category. They are the unanswerable class of Section III-D seen in the census, and five subtypes account for 39 of them. Ordinal questions (9) ask for a first, last, or smallest member. The relevant relation is explored and the candidates come back, but no component compares them. Asked for the year of the first Miss America pageant, the system answers 1925, then 1980, then 1929 across successive rounds, because nothing sorts and takes the extreme. Numeric literals (16) are internal identifiers and codes that CWQ's templates invoke and that no explored relation exposes. Date literals (4) and temporal qualifiers (4) fail because the environment holds events but not their dates. Asked who was president in 1988, the system names the winner of that November's election, who took office the following January. Only the last subtype, data errors (6), is a gap in this particular import. The other four are gaps in what a triple-and-traversal architecture can express at all. No larger graph fixes them, since they need an operator.

D. The Echo Attractor

Thirteen failures name a mechanism that recurs independently of any component. The system resolves a real, well-grounded entity that sits near the correct answer in the graph. The verifier passes it, because it genuinely is grounded. The fact is true but not what was asked. Two shapes appear. In the first, the system confuses entities in one tight cluster. Asked who plays Lex Luthor on *Smallville*, it answers that John Glover plays Lionel Luthor. In the second, it finds the right entity family at the wrong level. It answers "Kingdom of Denmark" where the gold answer is "Denmark", or narrows from Harvard College to Harvard University. None of these are cases where the system knew nothing. The right edge exists, is used, and lands one node away.

The echo attractor is also why the consensus pass of Section V-C flagged far more questions than adjudication confirmed. All five systems answer "rick scott" where the gold answer lists his professions, and "thor heyerdahl" where it lists his occupations. Each returns the topic entity. All five answer "belgium" where the gold answer is the containing region, and "maryland" where it lists counties. When systems agree with one another against gold, the usual cause is a shared attractor, not a label error. A policy of rescoring whenever three systems agree would have turned this failure into apparent correctness. Adjudication is what stands between the two. Counting the census and the earlier exclusions together, 57 distinct questions needed the benchmark, not the system, to be corrected. Two patterns account for most of the CWQ defects, and both are properties of template generation rather than of annotation care. One is a gold list flattened from a relation that has several values, with the pipeline keeping a marginal one. The other is an annotation query that traversed one relation too far.

VIII. Discussion

A. Findings against the Research Questions

RQ1. Agentic navigation improves multi-hop accuracy, and the multi-hop qualifier is doing real work. AGR leads every accuracy column, and all eight paired tests reject. On CWQ its accuracy rises with hop count, while every static system decays. Three boundaries come with that answer. Think-on-Graph hedges marginally less on CWQ, at 22.5% against 23.0%, the one cell of Table IX that AGR does not lead. It is also better on CWQ's one-hop stratum. And the margin over it is located in the budget, against a baseline that prunes from narrower candidate sets. So the claim defended here is that guided search completes under a realistic budget where beam search does not.

RQ2. The answer has three parts that a yes-or-no question would merge. First, navigation eliminates structural hallucination, and the verifier is not what does it. Second, the precision advantage over Think-on-Graph belongs to the architecture, and this experiment cannot award any of it to the layer. Removing the verifier moves per-question precision by $+0.001$ and -0.010, an order of magnitude below the gap. Entity filtering runs only on the give-up path, which 10 WebQSP and 29 CWQ questions entered. A filter that touches 2.5% of a test set cannot produce a 0.13 precision margin on it. Two differences that the ablation holds fixed remain live. One is the groundedness filter on evaluator resolutions, which is navigation rather than verification. The other is the drafting prompt's provisions, which the baselines' plain prompt lacks. Third, verification shows no measurable Hits@1 effect. A metric satisfied by one correct entity among several cannot see a mechanism that works on the others. What the layer offers is an answer that arrives with the triples supporting it. That is an auditability claim, not an accuracy claim.

RQ3. One condition of four reaches significance. Removing the planner improves WebQSP by 0.083 F1 at $p = 0.006$ while saving 31% of tokens, and it costs CWQ 0.047 F1 at $p = 0.088$. That asymmetry, not a pooled average, is the result. It says how decomposition should be applied, not whether it works. Backtracking, verification, and the scoring blend show no detectable effect at $n \approx 200$. On cost, the whole apparatus runs at half the calls of the agentic baseline.

So the three answers do not point the same way. Navigation is where the accuracy comes from, and verification is where the auditability comes from. Decomposition, the component the literature treats as straightforwardly beneficial, is the one this work can show is sometimes harmful.

B. What Agency Buys, and What It Costs

Agency buys the deep tail. On CWQ the agentic system's accuracy rises across hop strata while every static system decays. It buys that at roughly an order of magnitude more tokens than one-shot retrieval. Against the other agentic system, it buys higher accuracy for half the calls, and it never exhausts the call budget that clips that baseline on 44% of CWQ questions. That is the difference between a guided search and a blind one, priced. Agency also lengthens the failure surface. Decomposition errors, abandoned relations, and verifier errors are failures that a static baseline cannot have. It has no plan to mis-decompose, no frontier to abandon, and no draft to check. Agency converts a retrieval problem into a control problem, and control problems fail in more ways.

C. When Verification Helps and When It Over-Hedges

The layer's contribution is auditability, and its cost is recall. Both directions show up at case level, which is where they are established, since the aggregate effect of removing the layer is indistinguishable from zero. The layer helps when a draft asserts something the traversal never supported. It rejected the fabricated Brando claim of Section VII before it could be emitted. A pair of sibling questions about the battle of Vicksburg shows the other kind of help. One asks who was president during the battle, and its gold answer is Ulysses S. Grant, who took office six years later. AGR answered that the claim could not be verified, and its verifier flagged the draft. The ablated system without a planner asserted Grant with confidence and was scored a hit for restating a gold value that answers a different question. The system that hedged was the more correct one, and the metric scored it as a loss.

The layer over-hedges when a true claim has no single edge stating it in the drafted phrasing. The five wrongly rejected cases of Section VII show that, and four of them were retries that ran and lost the answer. So the claim is narrow. Verification does not make this system more accurate. It makes the answers it does give attributable, at a small and measurable cost in recall. And it provides no protection against an answer that is grounded, connected, true, and not what was asked.

D. Threats to Validity

The environment is friendlier than the knowledge base it stands in for. It is a union of subgraphs that were extracted to contain their own answers. That is why reachability is certified at 97.3% and 99.7%, and why the accuracies are not comparable with evaluations over full Freebase. Every system navigates the identical environment, so the relative ordering is sound even where the absolute level is optimistic. Every claim in this paper is relative for that reason.

An identical environment is not identical access. The agentic baseline sees narrower candidate sets than AGR. The static baseline sees a single logical hop under an unordered fanout cap. So the reported margins are bounded in both directions, and equal-width re-runs would settle them.

Every number comes from one run per condition. The intervals and paired tests resample questions, so they bound the sampling variance of the draw and nothing else. The backbone repeated its trajectory on only 8 of 12 probes. The margins defended here are large relative to their intervals, so a rank reversal from decoding noise alone is unlikely. But the experiment that would bound that variance was not run.

Entities are keyed by name. Homonyms collapse into one node. A merged node can manufacture a path between unconnected things, which inflates reachability and lets a wrong answer verify as grounded. So structural groundedness is a claim about this graph's node identity, not about the world.

Wrongful acceptance is unmeasured. The verifier logs the claims it rejects and not the ones it accepts. So one polarity of its error is reported at census scale and the other from a single specimen. The exposed population is bounded only as $[39, 2{,}008]$ of $2{,}008$ accepted claims.

Further limits. The judge fell short of its pre-specified bar by 0.0005 and shares the systems' backbone. Topic entities are given rather than linked, so the accuracies assume a linking step that a deployed system would perform itself. The ablations are underpowered. The extraction bug makes the measured accuracy a lower bound on one question shape. And the benchmark defects were adjudicated by the first author, with the consensus signal visible.

One criterion is not reported. Path fidelity asks whether a system reaches the answer by the relation chain that the

benchmark's own query specifies. It needs the gold SPARQL chains, and the distribution used here publishes triples without them. Reconstructing the chains would measure whether the right answer came by the right path. The groundedness metrics can only approximate that question, and it bears directly on the echo attractor.

IX. Conclusion

This paper asked what an agent gains by navigating a knowledge graph rather than retrieving from it. It also asked what a claim-level check against the traversed subgraph adds on top. AGR leads every accuracy column on WebQSP and CWQ, at half the language-model calls of the agentic baseline. Its margin over that baseline is affordability under a fixed budget rather than better resolution. On CWQ it is the only system whose accuracy rises with hop count. Structural hallucination is eliminated by navigation itself. The verification layer contributes auditability rather than accuracy, and decomposition helps or hurts depending on the structure of the question.

The failure census earns three small repairs, each backed by a repeated mechanism. The claim decomposer should be told which argument of the drafted sentence is the answer. The evaluator's decisions should be logged beside the scores of the relations it abandons. And the drafter should commit to a value that the evaluator resolved and the verifier grounded. One logging change has the highest value for its cost. Persisting each accepted claim with the triples that matched it would turn wrongful acceptance into a rate with a denominator. It would also make the output contract checkable after the run. Three further additions are specified by the failure categories. A decomposition step should be gated on whether its rewrite keeps the question's interrogative and its constraints. A set-intersection operator would serve the 47 composite-claim failures, and ordinal and literal-value support would serve the knowledge-graph gaps. A semantic check that accepts a bounded relation chain would recover the claims wrongly rejected for lacking a single edge. Finally, re-running the agentic baseline at AGR's candidate widths, and the static baseline at a wider and ordered radius, would lift the two baseline bounds.

Four directions extend the work rather than repairing it. A knowledge graph constructed by a language model, used as a controlled comparison, would separate what the framework contributes from what a curated environment contributes. Rollout-based value estimation would supply what the scorer lacks, since it scores each candidate once and never revises that score. Temporal and multi-source environments would address the questions with a time qualifier that recur through the census. And high-stakes domains are where auditability matters more than accuracy.

The result we would most want carried forward is not the accuracy margin. It is the shape of the gap that remains. A system that traverses a real graph and copies its answers out of real triples can be made to assert nothing ungrounded. What remains is an answer that is grounded, connected, and true, and is still not the answer to the question that was asked. Fact verification against a knowledge graph turns out to be the easier half of the problem. *Relevance* verification is the half still open.

Data Availability

Code, prompts, per-question run records, and the scripts that produce every table and figure in this paper are available at https://github.com/sakif-khan/agentic-graph-reasoning under the MIT licence. The questions and the Freebase-derived subgraphs come from the public Reasoning-on-Graphs distribution [7], used unmodified.

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